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Proposal of a general framework to categorize continuous predictor variables

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Abstract

The use of discretized variables in the development of prediction models is a common practice, in part because the decision-making process is more natural when it is based on rules created from segmented models. Although this practice is perhaps more common in medicine, it is extensible to any area of knowledge where a predictive model helps in decision-making. Therefore, providing researchers with a useful and valid categorization method could be a relevant issue when developing prediction models.

In this paper, we propose a new general methodology that can be applied to categorize a predictor variable in any regression model where the response variable belongs to the exponential family distribution. Furthermore, it can be applied in any multivariate context, allowing to categorize more than one continuous covariate simultaneously. In addition, a computationally very efficient method is proposed to obtain the optimal number of categories, based on a pseudo-BIC proposal. Several simulation studies have been conducted in which the efficiency of the method with respect to both the location and the number of estimated cut-off points is shown.

Finally, the categorization proposal has been illustrated in a real data set of 543 patients with chronic obstructive pulmonary disease (COPD) from Galdakao Hospital's five outpatient respiratory clinics, who were followed up for 10 years. Exercise capacity is known to be an important predictor of adverse events in patients with COPD. In this paper, we applied the proposed methodology to jointly categorize the continuous variables six-minute walking test and forced expiratory volume in one second in a multiple Poisson generalized additive model for the response variable rate of the number of hospital admissions by years of follow-up. The location and number of cut-off points obtained were clinically validated as being in line with the categorizations used in the literature.

Keywords: Cut-off point estimation; Pseudo-BIC, GAM, COPD.

1 Introduction

Many clinical predictors are recorded as continuous variables, but in practice, clinical guidelines and scoring systems used to predict patients' morbidity and mortality often require the expression of these as categorical variables. The development of clinical practice guidelines and scoring systems is necessary in order to facilitate clinicians' decision-making process and help them reduce variability in clinical practice.

In addition, in the development of prediction models for use in practice, continuous predictor variables are often categorized (Barrio et al., 2021; Spruit et al., 2012). These models provide estimates for individual risk of suffering some unfavorable event, which helps with decisions such as whether or not a given patient should be discharged, etc. Therefore, how to categorize a continuous variable into a categorical one is of great interest to clinical researchers, clinicians, and healthcare professionals in general.

When developing prediction models, the selection of the covariates (clinical variables) to be used in the model is essential, as well as good modeling of their relationship with the outcome. From a statistical point of view, categorizing continuous variables is not regarded as advisable, since it may entail a loss of information and power (Altman, 2005; Royston et al., 2006). Additionally, there are statistical modeling techniques such as generalized additive models (GAM) (Hastie and Tibshirani, 1986) which do not require any assumption of linearity between predictors and response variables, and so allow for the relationship between the predictor and the outcome to be modeled more appropriately. Nevertheless, one of the limitations that GAMs may have in practice is that it is not always easy to obtain decision rules based on this type of model, since the interpretability of GAMs may depend on the complexity of the risk functions (Hegselmann et al., 2020). In this case, another possibility to achieve a more interpretable non-linear effect is to discretize the covariates, this is, turn it into a categorical covariate. In fact, based on the results of a survey of the epidemiological literature, in the 86% of the papers included in the study, the primary continuous predictor was categorized, of which 78% used 3 to 5 categories (Turner et al., 2010).

The estimation of a single cut-off point to dichotomize a continuous variable, a biomarker or a risk score, is a topic that has been discussed over a long period of time. Miller and Siegmund (1982) proposed to use the maximum chi-squared statistic (or minimum p-value) as the criterion to estimate the cut-off point for dichotomization purposes. Altman et al. (1994) showed that the minimum p-value approach yields an increase in the type I error rate and thus proposed a correction for the minimum p-value formulae. Latter, Faraggi and Simon (1996) proposed a cross-validation approach to estimate the cut-off point based on the minimum p-value approach. On the other hand, Hin et al. (1999) proposed the use of GAM and the curves obtained from them to dichotomize continuous variables in clinical prediction models. In parallel, proposals have been developed for the estimation of optimal cut-off points based on different criteria related to the predictive capacity of the predictive scores or variables to be categorized. For example, by maximizing the sensitivity or specificity parameters (i.e., probability of classifying correctly an individual with/without the event of interest), Youden or Kappa indexes (Youden, 1950; Cohen, 1960; Greiner et al., 2000), among others. Some of these criteria have been implemented in the `OptimalCutpoints` `R` package (López-Ratón et al., 2014).

However, it may be preferable to use more than two categories, as this serves to reduce the loss of information and also allows the relationship between the covariate and the response variable to be better preserved. In this regard, the selection of more than one cut-off point is a more recent topic of research. This has been often done based on percentiles, even though this is known to have drawbacks (Bennette and Vickers, 2012). Barrio et al. (2013) proposed a categorization methodology using GAM with P-spline smoothers, in which they considered creating at least one average-risk category along with high- and low-risk categories based on the GAM smooth function. However, the need for an additional category relied on a subjective decision. On the other hand, in the context where the response variable takes two possible values, Tsuruta and Bax (2006) proposed a parametric method for obtaining cut-off points based on the overall discrimination statistic, c-index (Harrell et al., 1982). The authors showed the optimal location of cut-off points in a case where the distribution of the continuous predictor variable is known, yet, in practice, continuous predictor variables do not usually respond to a known distribution. In the same context, Barrio, Arostegui, Rodríguez-Álvarez and Quintana (2017) proposed a methodology to select the optimal cut-off points considering the maximal discrimination ability of the logistic regression model measured by the area under the receiver operative characteristic (ROC) curve (AUC), which for binomial response variables is equivalent to the c-index (Bamber, 1975). Unlike the Tsuruta and Bax (2006) proposal, the Barrio, Arostegui, Rodríguez-Álvarez and Quintana (2017) proposal does not require distribution assumptions and can be used in any univariate or multiple modeling situation regardless of the distribution of the original continuous predictor. Furthermore, this methodology has also been extended for use in the Cox proportional hazards regression model (Barrio, Rodríguez-Álvarez, Meira-Machado, Esteban and Arostegui, 2017), where different estimators for the concordance probability were considered in order to measure the discrimination ability of the categorical variable. Although this methodology allows selecting any possible number of cut-off points, either in a univariate or a multiple context, simultaneously categorizing more than one variable is computationally very expensive, and furthermore, in many circumstances the number of categories in which to categorize the predictor variable is unclear.

The optimal number of the cut-off points has been recently studied by Chang et al. (2019) and Barrio et al. (2021). The former considered the Cox proportional hazard regression model and proposed an AIC criterion to estimate the number of cut-off points, where the AIC values were corrected with cross-validation and Monte Carlo methods. On the other hand, the latter proposed a bootstrap-based hypothesis test in the logistic regression setting. In both cases, the proposed methodology has been developed for use in a specific regression model.

While several existing approaches have been proposed to categorize continuous variables, such as those by Barrio, Rodríguez-Álvarez, Meira-Machado, Esteban and Arostegui (2017) in the context of logistic regression and Barrio et al. (2021) using bootstrap-based hypothesis testing for selecting the optimal number of cut-off points, these are typically developed for specific types of regression models and are not easily extendable to broader contexts. In this work, we propose a broader approach which allows researchers to categorize continuous variables in a wider variety of regression problems. We propose fitting a single GAM in which we estimate smooth curves for each covariate that we intend to categorize. Building upon these estimates, we develop a methodology that approximates the smooth functions of GAMs through piecewise constant representations. This approach stands in contrast to that of Barrio, Rodríguez-Álvarez, Meira-Machado, Esteban and Arostegui (2017); Barrio et al. (2021), which identify cut-off points by maximizing the discriminatory ability of the categorized variable.

In addition, the methodologies proposed by Barrio, Arostegui, Rodríguez-Álvarez and Quintana (2017) and Barrio et al. (2021), although they can be applied in a multiple model, present computational limitations when it comes to categorizing more than one variable at the same time, or the categorization of a variable according to the different factors taken by another variable with which there is a significant interaction. Thus, the methodology presented in this paper gives a solution to the limitations of the previous methods, allowing first the categorization of continuous variables in a different context than that of the logistic regression, and second, a computationally very efficient method when categorizing one or several continuous variables in any multiple contexts. This proposal is based on the idea that the smooth function of the GAM model can be approximated by a piecewise function, and provides a global framework to select both the location and the number of optimal cut-off points for any response variable with a distribution of the exponential family, which in addition, can be applied for any kind of multiple regression modeling approaches, including interactions or smooth functions.

Other related approaches include change-point models and adaptive splines. Change-point models (Hawkins, 2001) are typically designed to detect abrupt shifts in the distribution or regression function, often under specific distributional assumptions, and their primary goal is to identify structural breaks rather than to provide interpretable categorizations. On the other hand, Goepp et al. (2025) proposed adaptive spline regression with automatic knot selection via an ℓ_1-type penalty, aiming at a parsimonious yet flexible approximation of the regression function. While these methods share with our framework the idea of controlling complexity, their objectives differ: change-point models focus on structural change detection, and adaptive splines focus on function approximation and prediction accuracy. By contrast, our proposal explicitly targets the approximation of smooth functions through meaningful cut-off points that yield interpretable categorizations of continuous covariates, thereby prioritizing parsimony and interpretability alongside model fit.

The rest of the paper is organized as follows. In Section 2 we introduce the main statistical models and notation that we will use in this document, starting with a description of the generalized linear and additive models (GLM, GAM). In Section 3 we describe the proposal presented in this work, whose goal is twofold. On the one hand, to select the optimal location of the cut-off points and, on the other, to select the number of categories in which to categorize the continuous variable. Section 4 outlines several simulation studies conducted with the aim of validating the proposed methodology as well as the criteria for selecting the number of cut-off points. We continue with the application to a real case study of patients with a chronic obstructive pulmonary disease which is presented in Section 5. Finally, the paper closes with a discussion in Section 6 in which the main findings are reviewed and conclusions are drawn.

2 Notation and preliminaries

In this Section, we introduce the general notation as well as the main statistical models we use throughout the paper.

The generalized linear model, (GLM), was introduced by Nelder and Wedderburn (1972) and further developed by McCullagh and Nelder (1989). This approach has three main differences with respect to the linear model, which are: a) the normal distribution for the response variable is replaced by the exponential family distribution, b) a monotonic link function $g(\cdot)$ is used for modeling the relationship between the expected value of the response variable and the covariates and finally c) it uses an iterative reweighed least squares algorithm for the parameter's estimation. The GLM model can be written as:

$$g(E(Y \mid \mathbf{X})) = \beta_0 + \beta_1 X_1 + \ldots + \beta_Q X_Q, \ (1)$$

for Y a response variable whose distribution belongs to the exponential family, $(X_1, \ldots, X_Q)$ a vector of Q covariates and $g(\cdot)$ the link function.

The GLM model assumes a linear relationship between the covariates $X_1, X_2, \ldots, X_Q$ and the link function $g(\cdot)$. However, in real life, effects are often not linear and we might be interested in a more flexible approach to model this relationship. This approach is called the generalized additive model (GAM), and was first introduced by Hastie and Tibshirani (1990). The GAM model can be written as:

$$g(E(Y \mid \mathbf{X})) = \beta_0 + f_1(X_1) + \ldots + f_Q(X_Q), \ (2)$$

where $f_j(), \quad j = 1, \ldots, Q$ stands for a non-parametric function applied to the continuous covariates (or some of). The idea is to let the data determine the relationship between the linear predictor $\eta = g(\mu(\mathbf{X}))$ and the explanatory variables rather than enforcing a linear relationship.

The original estimating method consisted of estimating the $f_j()$ by iterative smoothing of partial residuals with respect to the covariates. Later, the use of other methods such as kernel smoothers (Wand and Jones, 1994) or spline-based smoothers (Green and Silverman, 1994) were proposed. Nevertheless, the penalized spline regression (P-splines, Eilers and Marx (1996)) has become one of the most popular methods to estimate the smooth functions $f_j()$ (Eilers et al., 2015). The method is based on the representation of the smooth component as a combination of base functions and on the modification of the likelihood function by introducing a penalty based on differences between adjacent coefficients. In this context, the standard estimation method is the restricted maximum likelihood method (REML). However, in some circumstances, the penalties present an overlapping structure, with the same coefficients being penalized simultaneously by several smoothing parameters. In order to deal with this problem, Rodríguez-Álvarez et al. (2019) have proposed an adaptive fast estimation method.

However, despite being a very flexible model, and being interpretable, in practice the GAM may not always be helpful when it comes to using the model during the decision-making process. In that case, it is easier if approximated by piecewise constants as shown in Figure 1. The present work is based on this idea.

3 Methods

3.1 Proposal to select the optimal location of cut-off points

Let Y be a response variable belonging to the exponential family distribution, $X_1, \ldots, X_T$ a set of T continuous predictor variables which we want to categorize (without loss of generality we consider the first T in the set of Q covariates) and $(X_{T+1}, \ldots, X_Q)$ a vector of $Q-T$ predictor variables which can be either continuous or categorical variables. For ease of notation, we will consider them hereinafter as continuous variables.

If it is considered a vector of k_j cut-off points (ordered from lowest to highest) $\mathbf{c}_j = (c_{j,1}, \ldots, c_{j,k_j})$ which defines the $k_j + 1$ intervals for a covariate X_j ($j = 1, \ldots, T$) the effect $f_j(X_j)$ of the covariate X_j can be approximated by

$$f_j(X_j) \approx \beta_0 + \sum_{s=1}^{k_j} \beta_{j,s} I\left\{c_{j,s} < X_j \leq c_{j,s+1}\right\}, \quad \text{for} \quad j = 1, \ldots, T$$

where $\beta_{j,s}$, $j = 1, \ldots, T$ and $s = 1, \ldots, k_j$ are unknown coefficients. Note that we have considered $[min(X_j), c_{j,1}]$ as the reference category, and, for simplicity of notation, we consider that $c_{j,k_j+1} = \infty$, $j = 1, \ldots, T$.

Therefore, the GAM model in (2) can be approximated by the model in (3)

$$g(E(Y \mid \mathbf{X}, \mathbf{c})) = \beta_0 + \sum_{j=1}^{T} \sum_{s=1}^{k_j} \beta_{j,s} I\left\{c_{j,s} < X_j \leq c_{j,s+1}\right\} + \sum_{j=T+1}^{Q} f_j(X_j). \quad (3)$$

Under the model in equation (3) it is very easy to interpret the effect of the covariates X_j $j = 1, \ldots, T$ on the response and makes it possible to establish rules in a much simpler way than the GAM model presented in equation (2). In addition, it facilitates the stratification of individuals, adjusting for the rest of covariables X_j $j = T+1, \ldots, Q$, in a simpler way than with a GAM. It also allows comparisons to be made with other studies where categorizations have been used.

The vector of cut-off points $\mathbf{c}_j = (c_{j,1}, \ldots, c_{j,k_j})$ in equation (3) can be established beforehand, or an estimation will have to be obtained from the data. Note here that sometimes $\mathbf{c}_j$ may not be known but the value of k_j may be established a priori. In other cases both $\mathbf{c}_j$ and k_j may be unknown, for $j=1,\ldots,T$. We describe below our proposal for estimating both the location and the number of cutoff points when both are unknown.

Step 1

Given a sample $\{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$, we propose to estimate the GAM model in equation (2) using p-splines and an adaptive fast estimation algorithm (Rodríguez-Álvarez et al., 2019) to estimate the smooth functions $f_j(), \quad j=1,\ldots,Q$.

$$g(E(Y\mid\mathbf{X}))=\beta_0+\hat{f}_1(X_1)+\ldots+\hat{f}_T(X_T)+\hat{f}_{T+1}(X_{T+1})+\ldots+\hat{f}_Q(X_Q). \quad (4)$$

For $j=1,...,T$ **:**

Step 2

Obtain the estimates $\hat{f}_j(x_{j1}),\ldots,\hat{f}_j(x_{jn})$ and the standard errors $\hat{\sigma}_j(x_{j1}),\ldots,\hat{\sigma}_j(x_{jn})$, resulting from the fit of the model in (4).

Step 3

Given $\mathbf{c}_j$ we define the **weighted mean squared error**

$$WMSE(\mathbf{c}_j)=\sum_{i=1}^{n}\frac{\left(\hat{f}_j(x_{ji})-\overline{f_j}^{\mathbf{c}_j}(x_{ji})\right)^2}{\sigma_j^{\,2}(x_{ji})}$$

where

$$\overline{f_j}^{\mathbf{c}_j}(x_{ji})=\sum_{s=0}^{k_j}I\left\{c_{j,s}<x_{ji}\le c_{j,s+1}\right\}\left[\frac{\sum_{l=1}^{n}f_j(x_{jl})I\left\{c_{j,s}<x_{jl}\le c_{j,s+1}\right\}}{\sum_{l=1}^{n}I\left\{c_{j,s}<x_{jl}\le c_{j,s+1}\right\}}\right]$$

is the mean of the estimated $\hat{f}_j$ in the category $R_s^j=(c_{j,s},c_{j,s+1}]$ for $s=0,\ldots,k_j$.

Step 4

Estimate the vector of cut-off points $\mathbf{c}_j$

Initialize: Compute the initial estimates $(\hat{c}_{j,1}^0,\ldots,\hat{c}_{j,k_j}^0)$.

Step 4.1: Cycle $r=1,\ldots,k_j$ calculating the update

$$\hat{c}_{j,r} = \text{argmin}_c WMSE(\hat{c}_{j,1},\ldots,\hat{c}_{j,r-1},c,\hat{c}^0_{j,r+1},\ldots,\hat{c}^0_{j,k_j})$$

Step 4.2: Repeat **Step 4.1** replacing $(\hat{c}^0_{j,1},\ldots,\hat{c}^0_{j,k_j})$ by $(\hat{c}_{j,1},...,\hat{c}_{j,k_j})$ until the convergence is obtained.

3.2 Proposal to select the number of categories

Bayesian information criterion (BIC) is a useful solution to the problem posed by direct comparison of likelihoods when selecting a model (since the likelihood will never decrease even if redundant parameters are included in the model). It was originally proposed by Schwarz (1978) as a modification of Akaike's information criterion (AIC), by introducing a larger penalty term for the effective dimension of the model.

For any given model Θ, the BIC is defined as follows:

$$BIC = \phi log(n) - 2log(L(\Theta)) \quad (5)$$

where ϕ and $L(\Theta)$ stand for the effective dimension and likelihood of model Θ, respectively. Note that in the presence of estimated smooth functions f_j the effective dimension accounts for penalization, using the correction proposed by Wood et al. (2016).

Considering $\mathbf{c}_j = (\hat{c}_{j,1},\ldots,\hat{c}_{j,k_j})$ the vector of the optimal cut-off points for the covariate X_j $j=1,\ldots,k_j$ estimated on the basis of the method presented in Section 3.1 above (and a sample $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$), the estimated model for X_j $j=1,\ldots,T$ categorized based on these cut-off points would be,

$$g(E(Y \mid \mathbf{x},\mathbf{c})) = \hat{\beta}_0 + \sum_{j=1}^{T}\sum_{s=1}^{k_j}\hat{\beta}_{j,s} I\left\{\hat{c}_{j,s} < x_j \le \hat{c}_{j,s+1}\right\} + \sum_{j=T+1}^{Q}\hat{f}_j(x_j). \quad (6)$$

Thus, the BIC for the model in equation (6) would be

$$BIC^{\mathrm{c}} = \phi log(n) - 2log(L(\mathbf{c})), \quad (7)$$

with $L(\mathbf{c})$ and ϕ, the likelihood and effective dimension of the model represented in equation (6), respectively.

However, since in addition to model parameters ($\beta_{j,s}$ $j=1,\ldots,T$, $s=1,\ldots,k_j$) and smooth functions ($f_1,\ldots,f_Q$), also the cut-off points ($\mathbf{c}_j=(\hat{c}_{j,1},\ldots,\hat{c}_{j,k_j})$ $j=1,\ldots,T$) have been estimated, we define a pseudo-BIC, which considers the estimated number of cut-off points

$$BIC^{\mathbf{c}}_{pseudo}=BIC^{\mathbf{c}}+log(n)\left(\sum_{j=1}^{T}k_j\right). \quad (8)$$

We propose to use the minimum BIC_{pseudo} as the criteria to select the number of cut-off points for each variable X_j, $j=1,...,T$, by penalizing the effect of introducing an extra category (i.e, cut-off point) for the categorized variable. Let's denote $\mathbf{nc}=(nc_1,\ldots,nc_T)$ as the vector of the number of cut-off points needed to categorize the predictor variables $X_1,\ldots,X_T$, then:

$$\mathbf{nc}=argminBIC^{\mathbf{c}}_{pseudo}.$$

4 Simulation study

In this section, we present a simulation study with two different goals: 1) performance evaluation with regard to location, and 2) performance evaluation with regard to the selected number of cut-off points. Therefore, we conducted a simulation study considering a theoretically defined piece-wise multiple model for a gaussian, binomial and poisson response variables, considering different theoretical numbers of cut-off points for two continuous covariates we aimed to categorize.

The simulation study was performed in 64-bit R version 4.4.2 and run on a computer equipped with 16 GB of RAM, an Intel® Core™ i7-6700 processor, and running the Windows 10 x64 operating system.

Scenarios and set-up

Let, $\boldsymbol{X}=(X_1,X_2,X_3)$ be a vector of continuous covariates. We considered three different response variables: $Y_1\sim N(\mu(\boldsymbol{X}),\sigma=1)$, $Y_2\sim Bin(1,p(\boldsymbol{X}))$, and $Y_3\sim Po(\lambda(\boldsymbol{X}))$, with mean structures defined as follows:

$$\mu(\boldsymbol{X})=f_1(X_1)+f_2(X_2)+0.1X_3,(Gaussian) \quad (9)$$

$$[6pt]\log\left(\frac{p(\boldsymbol{X})}{1-p(\boldsymbol{X})}\right)=f_1(X_1)+f_2(X_2)+0.1X_3,(Binomial) \quad (10)$$

$$[6pt]\log\lambda(\boldsymbol{X})=\tfrac{1}{2}(f_1(X_1)+f_2(X_2)+0.1X_3),(Poisson) \quad (11)$$

For the Poisson case, $\lambda(\boldsymbol{X})$ was scaled to ensure that Y_3 followed a valid Poisson distribution. The continuous covariates X_1 and X_2 were drawn from a uniform distribution on $[-2,2]$, while X_3 was drawn from a uniform distribution on $[-1,1]$.

The functions $f_j(X_j)$, for $j=1,2$, were defined as piecewise-constant functions of the form

$$f_j(X_j)=\begin{cases} a_{j,1}, & X_j \le c_{j,1}, \\ a_{j,2}, & c_{j,1} < X_j \le c_{j,2}, \\ \vdots & \\ a_{j,k_j}, & c_{j,k_j-1} < X_j \le c_{j,k_j}, \\ a_{j,k_j+1}, & X_j > c_{j,k_j}, \end{cases}$$

where the vector of theoretical cut-off points $\boldsymbol{c}_j=(c_{j,1},\ldots,c_{j,k_j})$ (for $j=1,2$) was defined to obtain an equidistant sequence of values in the interval $[-2,2]$, leaving aside the extreme values.

We considered four scenarios defined by different theoretical numbers of cut-off points (k_1,k_2) for the covariates X_1 and X_2, respectively. These scenarios, together with the corresponding parameterizations of the piecewise-constant functions $f_1(X_1)$ and $f_2(X_2)$, are summarized in Table 1.

For sample sizes $n \in \{(500,1000,2000)\}$ and the four scenarios considered $m \in \{(S1,S2,S3,S4)\}$, $R=500$ independent samples $\{(\mathbf{x}_i, y_{1i}^m, y_{2i}^m, y_{3i}^m)\}_{i=1}^n$ were generated with $y_{1i}^m = N(\mu(\boldsymbol{X}=\mathbf{x}_i),\sigma=1)$, $y_{2i}^m = Bernoulli(P(\boldsymbol{X}=\mathbf{x}_i))$, and $y_{3i}^m = Po(\lambda(\boldsymbol{X}=\mathbf{x}_i))$. The performance of the proposed methodology to estimate the location of the cut-off points was evaluated by means of the bias and mean square error (MSE) of the estimated optimal cut-off points for each iteration as follows:

$$MSE_r = \frac{1}{k_1+k_2}\sum_{j=1}^{2}\sum_{s=1}^{k_j}(\hat{c}_{j,s}^r - c_{j,s})^2, \quad \text{for} \quad r=1,\ldots,R$$

where $\hat{c}_{j,s}^r$ is the estimated s^{th} optimal cut-off point in the simulation *r*, and $c_{j,s}$ is the s^{th} theoretical cut-off point, for the covariate X_j with $j=1,2$.

The number of cut-off points was selected considering those for which the BIC_{pseudo} was minimized.

Results

The numerical results obtained for the estimated optimal cut-off points for different sample sizes ($n \in \{500, 1000, 2000\}$), scenarios ($m \in \{S2, S4\}$) and $R = 500$ replicates are summarized in Table 2, Table 3 and Table 4, for response variables Gaussian, Binomial and Poisson, respectively. In addition, Figure 2 depicts the boxplots of the estimated location of the optimal cut-off points over 500 simulated data sets for a sample size of $n = 1000$. The results obtained for scenarios S1 and S3 are reported in the Supplementary Material (see Table S1, Table S2 and Table S3, for Gaussian, Binomial and Poisson response variables, respectively).

Simulation results suggest that, in general, the proposed methodology performs satisfactorily in regard to the estimation of the location of the cut-off points. The method performs best in the case of Gaussian response variables, while it behaves similarly in the case of binomial and Poisson distributions. As could be expected, as the sample size increases the bias and the MSE decrease. Nonetheless, the increase in the number of theoretical cut-off points does not significantly increase the MSE. For instance, in Scenario S1 (where $k_1 = 2$ and $k_2 = 1$) for a Binomial response variable, mean MSE values of 0.024 and 0.001 have been obtained for sample sizes of 500 and 2000, respectively (Table S2 in Supplementary Material). On the other hand, in scenario S4 (where $k_1 = 3$ and $k_2 = 2$) mean MSE values of 0.037 and 0.002 have been obtained for sample sizes of 500 and 2000, respectively (Table 3). Biases above 0.1 were only obtained in samples of size 500, in particular for a binomial response variable (never for a gaussian response variable), and in all cases, only at one of the cut-off points.

Regarding the number of cut-off points selected, Table 5 provides the percentage of simulations (out of 500) in which the correct combination of cut-off points has been selected based on the Pseudo-BIC criteria. In Table S4 in Supplementary Material the comparison with respect to the BIC criteria is provided. For the linear setting, the true number of cut-off points are selected in more than 80% of simulations, being this percentage higher for sample sizes of $n = 1000$ and $n = 2000$. However, for binomial and Poisson response variables, this accuracy rate is only maintained when sample sizes are 1000 or 2000, obtaining percentages close to 50% in scenarios S3 and S4 for $n = 500$. In Figure S3 in the Supplementary Material, we provide heatmaps representing the percentage of replicates in which each combination of cut-off points has been selected for a sample size of $n = 500$. The scenarios where the combination of cut-off points is correctly selected in only half of the replicates are S3 and S4, scenarios where the variable X_1 has three theoretical cut-off points. In general, the method tends to select fewer cut-off points, providing conservative results. This means that in terms of Pseudo-BIC, for a sample size of 500 and scenarios where one of the variables has 3 cut-off points, in terms of Pseudo BIC better results are obtained with fewer categories than the theoretical ones.

Computational efficiency

The computational cost of the proposed method lies in the search for the cut-off points, given that the smooth curve we want to approximate using a piecewise function is estimated only once. Therefore, we have compared computation times across different sample sizes and response variables, as well as for varying numbers of cut-off points, defined as the total number of cut-off points considered for variables X_1 and X_2. Figure 3 shows the mean, median and interquartile range of computational time by sample size and response variable in Scenario S4. Results obtained in Scenarios S1, S2 and S3 are available in Supplementary Material (see Figure S4, Figure S5 and Figure S6). The maximum number of cut-off points we have searched for in the simulation study are 12 (6 for X_1 and 6 for X_2). As expected, the computational time increases as the number of cut-off points increases, but in no case exceeds 15 seconds on average.

5 Application to patients with COPD

Chronic obstructive pulmonary disease (COPD) is considered a complex, heterogeneous, and multisystemic disease (Vanfleteren et al., 2016). Therefore, tools are needed to assess all aspects of the disease in a comprehensive and integrated approach. Hospital admissions and mortality are two of the key outcomes of COPD. To date, several models have been developed to predict adverse events in COPD patients (Esteban et al., 2008, 2022; Agusti et al., 2013). In this context, accurate assessment of exercise capacity (six-minute walk test) is considered an important variable due to the close relationship it has with patients' health and the development of adverse events (Spruit et al., 2012). In fact, the predictor variable six-minute walk test has been widely used in the creation of prognosis scales as BODE index (Celli et al., 2004), defined on the basis of four variables: Body Mass Index (BMI), Obstruction (forced expiratory volume in one second, FEV1), Dyspnea (MMRC scale), and Exercise (6MWT).

For illustrative purposes, in this paper, we show the usefulness and interest of the proposed methodology by applying it to a longitudinal database of patients with stable COPD, in order to jointly categorize the variables FEV1 and 6MWT in a multiple Poisson model. Specifically, in this study 543 patients being treated for COPD at five outpatient respiratory clinics affiliated with the Galdakao Hospital in Biscay between January 2003 and January 2004 were recruited. Patients were consecutively included in the study if they had been diagnosed with COPD for at least six months and had been receiving medical care at one of the hospital's respiratory outpatient facilities for at least six months. Their COPD had to be stable for six weeks before enrollment. Patients were assessed at baseline and at the end of the first, second, and fifth years, and were followed up for a total of ten years.

The data set was splited into Train (80%) and Test (20%) subsets. We considered the 10-year admission rate per number of years of follow-up as the response variable, for which a Poisson-GAM model was considered. The predictor variables 6MWT and FEV1, measured at the beginning of the study, were adjusted by means of a smooth function estimated using the adaptive fast estimation method using the `sop` function of the `R` package `SOP`, in the presence of a set of statistically significant baseline covariates such as BMI, Charlson index, age, quadriceps or shoulder strength and the number of previous hospital admissions in the Train set. The estimated relationship between the predictor variables 6MWT and FEV1 and the log scale of the 10-year admission rate in the multivariate Poisson GAM model can be observed in Figure 4. Both variables were categorized considering from 1 to 3 cut-off points. We got that the best categorization based on the minimum pseudo BIC obtained was to consider a unique cut-off point for both 6MWT and FEV1 predictor variables (pseudo-BIC=1246.59, BIC = 1234.50, and continuous model BIC = 1204.91). Details regarding the pseudo-BIC and BIC values for all the cut-off point combinations are available in the Supplementary Material (Table S8). We compared the performance of the categorized model against the original smooth model by evaluating the MAE and RMSE in both train and test samples. As can be seen in Table 6, the categorized version outperforms the continuous version in terms of fit metrics in the test sample.

The estimated cut-off points in the Train sample were 573 meters and 49.1 expiratory volume in percentage, respectively. These cut-off points are in line with the ones used previously in the literature (Celli et al., 2004; Vogelmeier et al., 2017).

The application of the proposed methodology to data from COPD patients with 10-year follow-up shows the usefulness and clinical practicality of the methodology, obtaining clinically validated results. Furthermore, this methodology allows us, unlike other existing categorization proposals, to obtain cut-off points simultaneously for more than one variable at the same time, and in the context of a Poisson model for rates.

6 Discussion

Although different methods have been proposed to select the optimal location of the cut-off points to categorize (or dichotomize) a continuous predictor variable (Hin et al., 1999; Tsuruta and Bax, 2006; Barrio et al., 2013; Barrio, Arostegui, Rodríguez-Álvarez and Quintana, 2017; Barrio, Rodríguez-Álvarez, Meira-Machado, Esteban and Arostegui, 2017) these approaches are generally developed within specific regression frameworks. To the best of our knowledge, up to now, no proposal has been made for the selection of the location nor the number of cut-off points in the broader context of GAMs or GLMs with exponential response distributions. Therefore, in this paper, we have presented a new methodological proposal that allows us to respond to this need and which turns out to be computationally very efficient. Different simulation studies have shown that this new proposal performs satisfactorily in regard to the location of the cut-off points and the selection of the number of categories, at least when we have a sufficient sample size to create the required number of categories. However that performance can decline for smaller samples, especially in the logistic and Poisson settings. This reflects the difficulty of estimating the smooth latent function from noisy piecewise data and the increased instability that arises when many cut points are present and some categories contain few observations. Thus, while the method works well for moderate-to-large samples, its use with smaller samples warrants caution.

Compared to previous proposals such as those by Barrio, Rodríguez-Álvarez, Meira-Machado, Esteban and Arostegui (2017); Barrio et al. (2021), our method stands out in several key aspects. First, while existing methods are tailored to specific models (e.g., logistic or Cox regression), our approach generalizes to any GAM setting involving exponential family distributions. Second, our algorithm is designed to simultaneously categorize multiple continuous covariates while being computationally efficient. Third, instead of relying on formal hypothesis testing for determining the number of categories, we use a novel pseudo-BIC criterion that balances model fit and parsimony. Taken together, these characteristics make our approach more versatile, scalable, and suitable for practical implementation in complex modeling situations where interpretability is a key aspect.

Notwithstanding these advantages, it is essential to acknowledge the risks and limitations of categorization itself. Transforming continuous predictors into categorical variables inevitably entails a loss of information, reduced statistical power, and the imposition of artificial thresholds that may not reflect underlying biological or clinical mechanisms (Altman, 2005; Royston et al., 2006). These issues are not eliminated by our method. Rather, the proposed methodology seeks to mitigate them by selecting cut-off points in a data-driven manner that reflects the smooth functional form estimated by the GAM, while prioritizing interpretability.

Therefore, it is important to note that the method is not a substitute for continuous modeling. Its application should be carefully motivated by substantive considerations, most notably, contexts where interpretability, decision-making, or clinical utility outweigh small losses in predictive accuracy. Misuse could arise if the method were applied mechanically, without regard for whether cut-off points are substantively clinically meaningful or validated in independent datasets. For this reason, we strongly emphasize the need for contextual validation: estimated cut-off points should be interpreted with input from domain experts, rather than treated as purely statistical artifacts.

With respect to the location of the cut-off points, we have seen that this method has a similar performance to previous proposals in the context of logistic regression and when categorizing a unique continuous variable (see Section B in Supplementary Material for details). However, the comparison with respect to the number of cut-off points is not trivial. The hypothesis testing proposed by Barrio et al. (2021) is computationally intractable when the objective is to categorize more than one variable simultaneously. Therefore, we have proposed a modified BIC criterion, similar to a variable selection process criterion, but penalizing for the number of cut-off points that are estimated. Computationally, the Pseudo-BIC criterion is significantly faster, allowing us to compare a large number of models (combinations of the number of categories) simultaneously. However, unlike a hypothesis test, we cannot speak of type I errors and power rates. Based on the results obtained in the simulation study, we can say that the pseudo-BIC criterion is a conservative criterion, particularly when the available sample size is small. However, despite the fact that this implies selecting fewer cut-off points than a priori necessary, we consider that it provides us with more parsimonious models. Finally, although the $log(n)$ penalty per estimated cut-point reflects the idea that each estimated cut-point contributes to model complexity, we acknowledge that a complete theoretical treatment of this adjustment, including the possibility of overlap with smoothing-based measures of effective dimension, is beyond the scope of this work. Therefore, the criterion should be viewed as primarily motivated by general model-selection principles and supported by its empirical performance in our simulations rather than by a formal asymptotic derivation. Moreover, as our simulation results show, the stability of the selected number of cut-points may decrease for smaller samples (e.g., $n = 500$), therefore we recommend to apply the method with caution in such settings.

The methodology presented in this paper was applied to a COPD patient's data set where we categorized two continuous covariates, 6MWT and FEV1, simultaneously in a Poisson regression model. These two variables have been widely used in predictive models to predict the evolution of COPD patients (Agarwala and Salzman, 2020). An example of this is the BODE index (Celli et al., 2004), where these variables are categorized in the creation of the score. The search for cut-off points for variables such as 6MWT is of current interest, but it is common to search for a single cut-off point (Spruit et al., 2012). The methodological approach allows us to find cut-off points for several variables simultaneously, obtaining cut-off points in a robust way. This facilitates the development of decision rules that are more transparent and easier to implement in practice, a necessary tool for clinical guidelines and bedside decision making.

We compared the original GAM model, which uses smooth functions for continuous variables, with the categorized model obtained via our proposed methodology. As shown in Section 5, the categorized model had slightly higher BIC than the continuous GAM model, indicating a minor trade-off in terms of fit. However, this reduction in model performance was relatively modest. In particular, when we applied both the original GAM model and the categorized model into a test sample, MAE and RMSE were smaller for the categorized model. Importantly, the categorized model provides a set of interpretable risk strata based on clinically meaningful cut-off points for 6MWT and FEV1, which are aligned with previously used clinical cut-offs. Therefore, while there is a measurable decrease in predictive accuracy, it is offset by a significant gain in model interpretability and clinical utility. This exemplifies the classic trade-off between performance and usability, and highlights the importance of choosing modeling strategies that align with the goals and context of the analysis. To the best of our knowledge, cut-off points for these variables have not been sought jointly in previous studies and neither in the specific context of a Poisson model. In addition, the results obtained, in particular those for the FEV1 variable, are in line with the cut-off points used so far in the literature.

An interesting topic for future research is the extension of the proposed methodology beyond additive structures to settings involving bivariate or multivariate smooths (e.g., tensor-product surfaces), which would allow for the categorization of interactions between continuous covariates.

In summary, we consider that the existence of a valid statistical methodology to categorize continuous variables is of great importance to support researchers in other areas. In many cases, continuous variables are categorized without a clear criterion (equidistant intervals or quartiles are often used). The methodology proposed in this paper ensures that the cut-off points as well as the categories obtained offer a model whose results in terms of goodness of fit and discrimination ability are very close to those obtained with the GAM model for the continuous variable, and also offer a clear advantage in terms of practical application and interpretability.

Disclosure statement

The authors have no conflicts of interest.

Data Availability Statement

The COPD dataset analyzed during the current study is not publicly available due to the sensible nature of the information and the associated privacy issues. The code for the methods presented in this work, is freely available at https://github.com/ibarrioberaza/SimultCutPoints.git.

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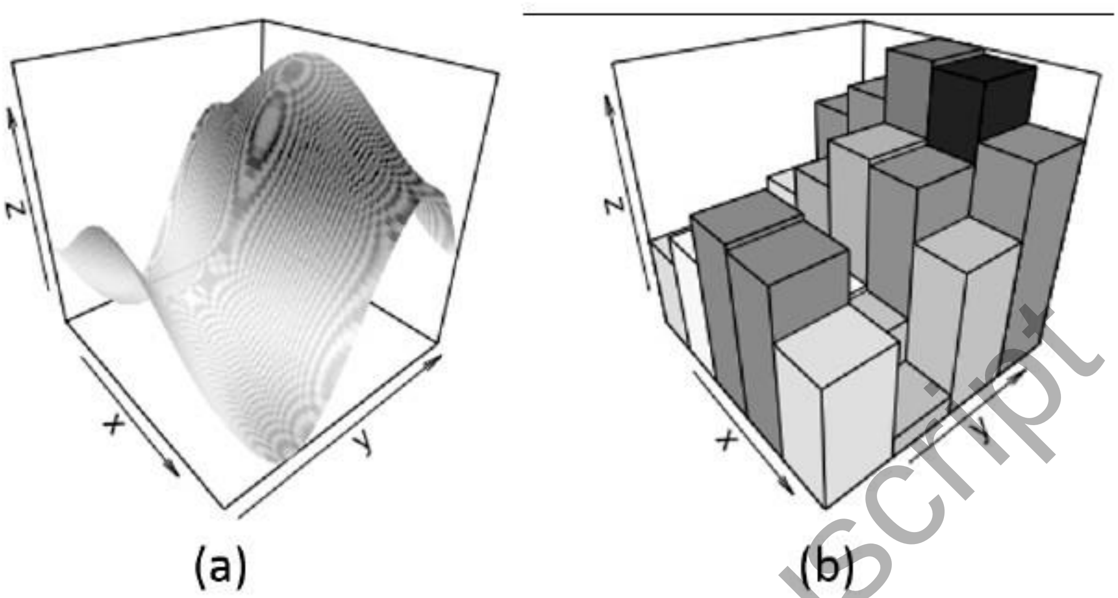
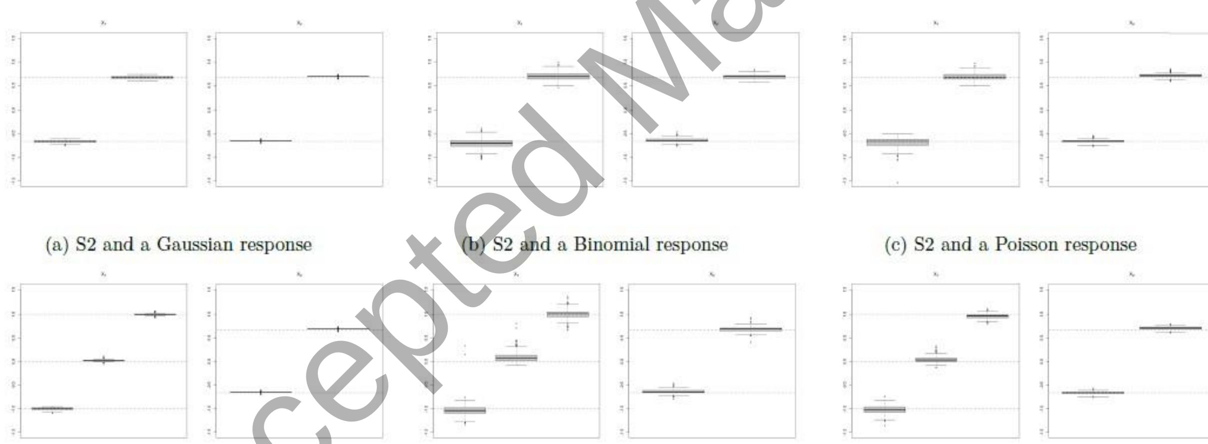


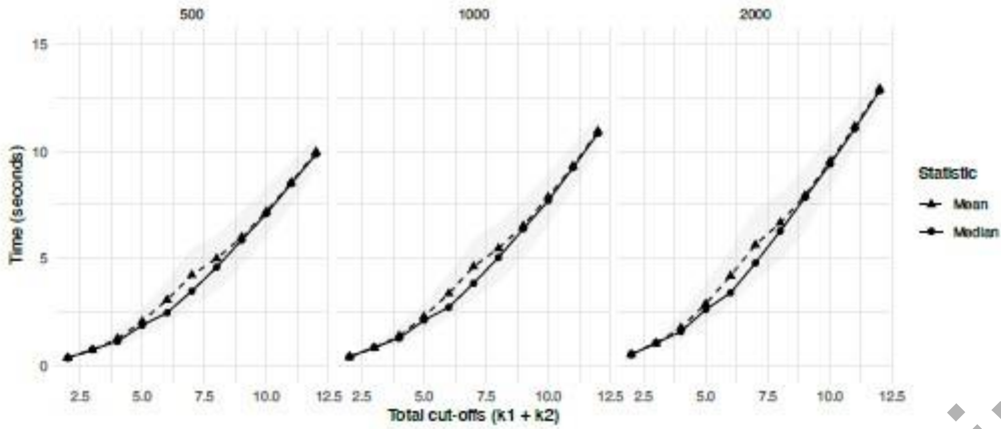
Figure 1: Illustrative example. a) Bivariate smooth relationship and b) Discretized relationship.



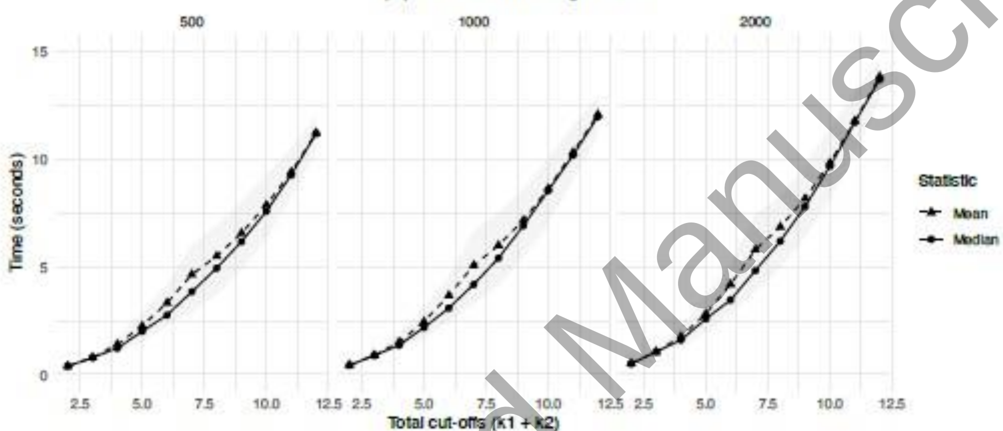
(a) S2 and a Gaussian response (b) S2 and a Binomial response (c) S2 and a Poisson response

(d) S4 and a Gaussian response (e) S4 and a Binomial response (f) S4 and a Poisson response

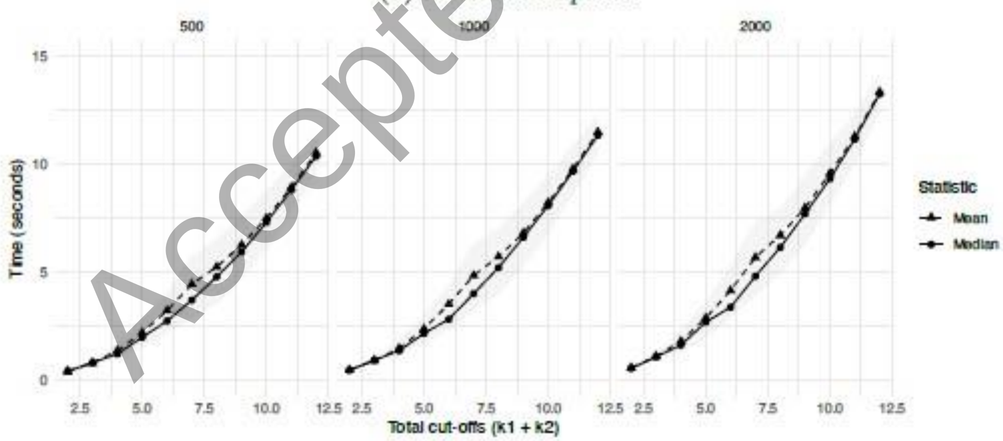
Figure 2: Boxplots of the estimated cut-off points for a sample size of $n = 1000$ in scenarios S2 (first row) and S4 (second row), for response variables Gaussian, Binomial, and Poisson, in first, second, and third columns, respectively



(a) Gaussian response



(b) Binomial response



(c) Poisson response

Figure 3: Mean, median and interquartile range (shaded area) of the time (in seconds) needed to search $k_1 + k_2$ number of cut off points in Scenario S4 by sample size. From top to bottom: gaussian, binomial and Poisson response variables.

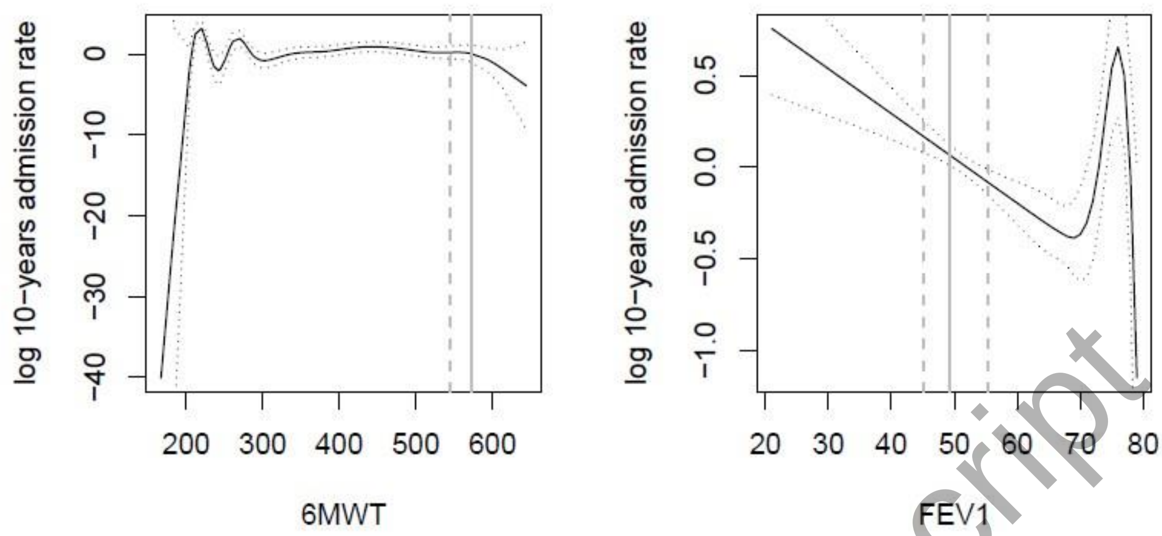


Figure 4: Estimated relationship between the predictor variables 6MWT and FEV1 and the log scale of 10-year admission rate in the multivariate Poisson GAM model. The solid line represents the location for $k=1$ cut-off point, while the dotted line represents the location for $k=2$ cut-off points. $k=1$ was selected as the optimal number of cut-off points for both 6MWT and FEV1, being this option the one with the minimum Pseudo BIC value, i.e., 1246.59

Table 1: Simulation scenarios. (a) Number of theoretical cut-off points for each covariate, (k_1, k_2), defining the four scenarios S1 to S4. (b) Parameterizations used for the piecewise-constant functions $f_1(X_1)$ and $f_2(X_2)$ in each case, where $\boldsymbol{a}_j$ denotes the function levels and $\boldsymbol{c}_j$ the corresponding cut-off points for X_j, $j = 1, 2$.

Scenario	k_1	k_2
S1	2	1
S2	2	2
S3	3	1
S4	3	2

(a) Numbers of cut-off points.

$k_1 = 2$		$k_1 = 3$		$k_2 = 1$		$k_2 = 2$	
$\boldsymbol{a}_1$	$\boldsymbol{c}_1$	$\boldsymbol{a}_1$	$\boldsymbol{c}_1$	$\boldsymbol{a}_2$	$\boldsymbol{c}_2$	$\boldsymbol{a}_2$	$\boldsymbol{c}_2$
1.5	−0.67	1.5	−1	−2	0	−2	−0.67
0	0.67	0	0	0		0	0.67
1.5		1.5	1			2	
		3					

(b) Parameterizations of f_1 and f_2.

Table 2: Numerical results for the location of the estimated optimal cut-off points for scenarios $m \in \{S2, S4\}$ and sample sizes $n \in \{500, 1000, 2000\}$ for a Gaussian response variable.

Scenario	**sample size**	**Variable**	**truec**	**Estimated cut-off point**			**MSE**	
				Mean	**Sd**	**Bias**	**Mean**	**Sd**
S2	500	X_1	-0.667	-0.688	0.046	-0.022	0.002	0.001
			0.667	0.644	0.040	-0.023		
		X_2	-0.667	-0.672	0.025	-0.006		
			0.667	0.684	0.021	0.018		
	1000	X_1	-0.667	-0.674	0.028	-0.007	0.001	0.000
			0.667	0.671	0.024	0.005		
		X_2	-0.667	-0.646	0.016	0.020		
			0.667	0.682	0.013	0.015		
	2000	X_1	-0.667	-0.664	0.019	0.003	0.000	0.000
			0.667	0.676	0.017	0.010		
		X_2	-0.667	-0.658	0.013	0.009		
			0.667	0.675	0.013	0.008		
S4	500	X_1	-1.000	-0.991	0.050	0.009	0.002	0.001
			0.000	0.029	0.033	0.029		
			1.000	0.974	0.044	-0.026		
		X_2	-0.667	-0.669	0.023	-0.002		
			0.667	0.688	0.021	0.021		
	1000	X_1	-1.000	-1.009	0.029	-0.009	0.001	0.000
			0.000	0.019	0.022	0.019		
			1.000	0.996	0.022	-0.004		

		X_2	-0.667	-0.648	0.016	0.019		
			0.667	0.684	0.013	0.017		
	2000	X_1	-1.000	-1.002	0.015	-0.002	0.000	0.000
			0.000	0.008	0.011	0.008		
			1.000	0.994	0.016	-0.006		
		X_2	-0.667	-0.657	0.013	0.010		
			0.667	0.674	0.013	0.007		

Table 3: Numerical results for the location of the estimated optimal cut-off points for scenarios $m \in \{S2, S4\}$ and sample sizes $n \in \{500, 1000, 2000\}$ for a Binomial response variable.

Scenario	sample size	Variable	truec	Estimated cut-off point			MSE	
				Mean	**Sd**	**Bias**	**Mean**	**Sd**
S2	500	X_1	-0.667	-0.801	0.171	-0.135	0.023	0.020
			0.667	0.716	0.144	0.049		
		X_2	-0.667	-0.707	0.099	-0.041		
			0.667	0.718	0.083	0.051		
	1000	X_1	-0.667	-0.717	0.113	-0.050	0.007	0.007
			0.667	0.690	0.076	0.023		
		X_2	-0.667	-0.643	0.046	0.024		
			0.667	0.682	0.043	0.015		
	2000	X_1	-0.667	-0.673	0.053	-0.006	0.002	0.002
			0.667	0.675	0.048	0.008		
		X_2	-0.667	-0.662	0.030	0.005		
			0.667	0.679	0.029	0.012		
S4	500	X_1	-1.000	-1.039	0.281	-0.039	0.037	0.088
			0.000	0.168	0.178	0.168		
			1.000	0.971	0.141	-0.029		
		X_2	-0.667	-0.714	0.114	-0.048		
			0.667	0.721	0.086	0.054		
	1000	X_1	-1.000	-1.047	0.121	-0.047	0.010	0.030
			0.000	0.089	0.110	0.089		
			1.000	0.998	0.089	-0.002		

		X_2	-0.667	-0.639	0.046	0.028		
			0.667	0.676	0.050	0.009		
	2000	X_1	-1.000	-1.013	0.047	-0.013	0.002	0.002
			0.000	0.029	0.050	0.029		
			1.000	1.000	0.044	0.000		
		X_2	-0.667	-0.651	0.034	0.015		
			0.667	0.676	0.033	0.009		

Table 4: Numerical results for the location of the estimated optimal cut-off points for scenarios $m \in \{S2, S4\}$ and sample sizes $n \in \{500, 1000, 2000\}$ for a Poisson response variable.

Scenario	sample size	Variable	truec	Estimated cut-off point			MSE	
				Mean	Sd	Bias	Mean	Sd
S2	500	X_1	-0.667	-0.744	0.160	-0.078	0.018	0.017
			0.667	0.683	0.134	0.016		
		X_2	-0.667	-0.724	0.088	-0.058		
			0.667	0.736	0.071	0.070		
	1000	X_1	-0.667	-0.688	0.107	-0.021	0.005	0.010
			0.667	0.676	0.070	0.009		
		X_2	-0.667	-0.663	0.039	0.004		
			0.667	0.705	0.037	0.038		
	2000	X_1	-0.667	-0.657	0.045	0.010	0.002	0.001
			0.667	0.664	0.041	-0.003		
		X_2	-0.667	-0.688	0.029	-0.022		
			0.667	0.699	0.025	0.033		
S4	500	X_1	-1.000	-1.031	0.210	-0.031	0.022	0.065
			0.000	0.125	0.163	0.125		
			1.000	0.946	0.095	-0.054		
		X_2	-0.667	-0.705	0.062	-0.038		
			0.667	0.724	0.050	0.057		
	1000	X_1	-1.000	-1.021	0.085	-0.021	0.004	0.004
			0.000	0.044	0.064	0.044		
			1.000	0.960	0.043	-0.040		

		X_2	-0.667	-0.667	0.031	0.000		
			0.667	0.704	0.029	0.037		
	2000	X_1	-1.000	-0.995	0.038	0.005	0.001	0.001
			0.000	0.015	0.038	0.015		
			1.000	0.971	0.024	-0.029		
		X_2	-0.667	-0.676	0.025	-0.010		
			0.667	0.693	0.019	0.026		

Table 5: Percentage of replicates (out of 500) in which the correct combination of cut-off points has been selected based on the Pseudo-BIC criteria for sample sizes $n \in \{500, 1000, 2000\}$, scenarios $m \in \{S1, S2, S3, S4\}$ and the three models considered (linear, logistic and Poisson).

Scenario	Model	Sample Size		
		n=500	n=1000	n=2000
S1	Linear	90.8	93.8	94.4
	Logistic	81.4	92	94.6
	Poisson	90.4	91.4	93.6
S2	Linear	87.2	92.6	88.8
	Logistic	75.2	87.2	89.6
	Poisson	77.6	85.6	86.4
S3	Linear	89	90.4	96.2
	Logistic	55	87.2	92.8
	Poisson	81	91.4	91.2
S4	Linear	82.8	89.6	90.4
	Logistic	52	84	91.2
	Poisson	64.2	80.8	84

Table 6: Fit metrics corresponding to the categorical and smooth models in train and test samples.

Subset	Model	MAE	RMSE
Train	Categorized	1.046	1.818
	Continuous	0.925	1.566
Test	Categorized	1.003	1.485
	Continuous	1.094	2.237.